

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

A possible reason for the website's connection timeout is a DoS attack. The logs show the server stops responding when it gets too many SYN requests. This could be a type of DoS attack called SYN flooding.

Section 2: Explain how the attack is causing the website to malfunction

When visitors try to connect to the website, a TCP three-way handshake happens:

1. The visitor sends a SYN packet to request a connection.
2. The server replies with a SYN-ACK packet and reserves resources.
3. The visitor sends an ACK packet to confirm the connection.

In a SYN flood attack, an attacker sends many SYN packets at once, using up the server's resources. This leaves no resources for real visitors, causing connection timeouts. The logs show the server is overwhelmed and can't handle new connections.